# ReviewFlow

## Complete Features Guide

A-Z Reference for All Modules, Conditions & Patterns

Laravel 12 | PHP 8.2 | MySQL | Sanctum API | Gemini AI

Razorpay Payments | Google Business Profile | Meta Graph API

30 Feature Sections | 24 Database Models | 106+ API Routes

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Use this document to replicate any feature/module in another project.

# Table of Contents


**1.** Architecture & Multi-Tenancy
**2.** Authentication & Authorization
**3.** Role & Permission System
**4.** Plan & Subscription System
**5.** Credit System (Pay-per-use AI)
**6.** Payment Integration (Razorpay)
**7.** Google Business Profile (GBP)
**8.** Review Management
**9.** AI Service (Gemini)
**10.** GBP Content (Posts & Photos)
**11.** Social Media Publishing
**12.** Lead Management (CRM)
**13.** WhatsApp Integration
**14.** Ad Reports
**15.** Business Insights
**16.** Google Audit
**17.** Competitor Analysis
**18.** Local Rank Checker
**19.** Keyword Generator
**20.** One-Click Optimization
**21.** AI Chat Mode
**22.** AI Media Generation
**23.** Invoicing & Accounting
**24.** Team Management
**25.** Admin Panel (Super Admin)
**26.** Health Score Dashboard
**27.** Email Verification
**28.** Mobile API (Flutter)
**29.** Environment Variables
**30.** Database Schema (Models)

**Appendix:** Route Groups | Key Patterns to Replicate

# 1. Architecture & Multi-Tenancy

## Tenant Hierarchy

```
Agency (top-level tenant)
Users (multiple, each has a role)
Clients (multiple businesses under one agency)
- GbpLocations (Google Business locations)
- Reviews, GbpPosts, GbpPhotos
- Leads, Invoices, Integrations (Google/Meta OAuth)
Subscription (one per agency - plan, credits, status)
SocialPosts, WhatsappMessages, AdReports, CreditLedger
```

## Scoping Pattern

Every query is scoped by **agency_id** from the logged-in user. Some users also have a **client_id** which further restricts their view to a single client.

```
$agencyId = $request->user()->agency_id;
Client::where('agency_id', $agencyId)->get();
```

## Key Models

| Model | Key Fields |
|---|---|
| Agency | name, slug, white_label (JSON) |
| User | agency_id, client_id, name, email, password, role, phone, avatar |
| Client | agency_id, name, industry, phone, email, gstin, billing_address |
| Subscription | agency_id, plan, credit_balance, monthly_credits, credits_reset_at, status, auto_reply, wa_notify |

# 2. Authentication & Authorization

## Web Auth (Session-based)

- **Register:** Creates Agency + Client + User + Subscription (TRIALING). Redirects to plans page.
- **Login:** Session-based. Admins go to /admin, trial users go to /plans, active users go to /dashboard.
- **Google OAuth Login:** Via Laravel Socialite (google_login driver). Auto-creates agency for new users.
- **Logout:** Invalidates session + CSRF token.

## API Auth (Token-based - Sanctum)

- **POST /api/login** - returns { token, user }, token via createToken('mobile')
- **POST /api/register** - creates agency + user + returns token
- **POST /api/auth/google** - Google ID token verification for mobile (throttled 10/min)
- **POST /api/logout** - revokes current token

- All authenticated routes use **auth:sanctum** middleware

### Mobile Login Bridge

**GET /mobile-login/{code}** - one-time code consumed to create web session from mobile app.

### Conditions

- Email uniqueness enforced on register
- Password minimum 8 characters
- Google OAuth: existing email = login; new email = auto-register
- Registration creates a TRIALING subscription (must upgrade to unlock AI features)

## 3. Role & Permission System

### Roles

| Role | Access |
|---|---|
| SUPER_ADMIN | Full admin panel, all agencies, bypass plan check |
| CLIENT_OWNER | Full access to their agency's data, manage team |
| CLIENT_STAFF | Limited to their assigned client's data |

### Middleware

**EnsureRole** (role:SUPER_ADMIN): Checks user->role against allowed roles. Returns 403 if not in allowed list.

**EnsureActivePlan** (plan): Bypasses for SUPER_ADMIN. Checks subscription status != TRIALING. Returns 402 JSON or redirects to /plans.

**Registered in bootstrap/app.php:**

```
$middleware->alias([ 'role' => EnsureRole::class, 'plan' => EnsureActivePlan::class ]);
```

## 4. Plan & Subscription System

### Plan Model Fields

```
name, code, price, gst_rate, credits, features (JSON), permissions (JSON), is_active, sort
```

### 12 Module Permissions

```
reviews, gbp_content, ai_media, audit, competitors, rank_checker, social, whatsapp, leads, keywords,
invoicing, ai_mode
```

### Per-Plan Credits

| Plan | Monthly Credits |
|---|---|
| STARTER | 100 |
| GROWTH | 500 |
| AGENCY | 2,000 |

## Subscription Flow

1. User registers - gets TRIALING subscription
2. User visits /plans - sees available plans with prices
3. User clicks upgrade - Razorpay checkout - payment verified - subscription activated
4. Monthly credits allocated based on plan tier

## GST Handling (India-specific)

All prices are GST-inclusive. Base and GST amounts computed on the fly:

```
$base = $price / (1 + $gst_rate / 100); $gst = $price - $base;
```

# 5. Credit System (Pay-per-use AI)

## Credit Cost Table

| AI Action | Credits | Description |
|---|---|---|
| ai_review_reply | 1 | AI-generated review reply |
| ai_social_caption | 1 | Social media caption |
| ai_keyword_gen | 2 | SEO keyword generation |
| ai_rank_check | 1 | Local rank check |
| ai_media_generate | 5 | AI image generation (most expensive) |
| ai_chat | 1 | AI marketing chat message |
| ai_post_caption | 1 | GBP post caption |
| ai_photo_caption | 1 | GBP photo caption |
| ai_optimize | 3 | One-click optimization |
| ai_audit | 2 | Google business audit |
| ai_competitor | 2 | Competitor analysis |

## Credit Gating Pattern

**canAfford()** - check balance before work
**Do the AI work...**
**deduct()** - deduct after success (transactional, row-locked)

Deduct is wrapped in DB::transaction() with lockForUpdate() on the Subscription row. Every change is logged in CreditLedger (audit trail).

### Top-up Methods

- **Monthly reset:** Plan-based, resets to plan's credit allocation
- **Manual top-up:** Admin adds credits via admin panel
- **Credit packages:** User buys a credit pack via Razorpay

## 6. Payment Integration (Razorpay)

No SDK - direct REST API calls with Basic Auth. Currency: INR.

### Payment Flow

1. **createOrder(amountRupees, receipt)** - creates Razorpay order. Amount x 100 = paise. Auto-capture enabled.
2. **Frontend:** Razorpay Checkout popup opens with order_id
3. **verifySignature(orderId, paymentId, signature)** - HMAC_SHA256 verification with hash_equals()

### Payment Model Fields

```
agency_id, plan, amount, currency, razorpay_order_id, razorpay_payment_id, status
```

## 7. Google Business Profile (GBP)

### OAuth Flow

1. **GET /clients/{client}/google/connect** - generates random state (stored in Cache, not session), redirects to Google consent. Scopes: business.manage, openid, email, profile.
2. **GET /google/callback** - exchanges code for access_token + refresh_token, stores in Integration (GOOGLE_GBP), fetches locations.

### Service Architecture

**GbpService** (facade) - GoogleGbpProvider (real API) or MockGbpProvider (fake data). Controllers only talk to GbpService.

### Capabilities

- **listReviews()** - fetch reviews from Google
- **putReply()** - reply to a review on Google
- **createPost()** - publish a post to Google
- **uploadPhoto()** - publish a photo to Google
- **fetchPerformanceMetrics()** - get business insights

### Conditions

- OAuth callback is OUTSIDE auth middleware (session can drop during redirect)

- State stored in Cache, not session
- Email verification required before connecting
- Disconnect removes the Integration record

# 8. Review Management

## Features

- **List reviews:** Per location, filterable, agency-scoped
- **Sync reviews:** Pulls latest from Google via GBP API
- **Reply to review:** Send reply through Google API
- **AI-generated reply:** Gemini generates a draft (credit-gated, 1 credit)
- **Draft reply:** Save AI reply as draft before sending
- **Sentiment analysis:** Local keyword matching (no AI call, no cost)
- **Auto-reply toggle:** Per-subscription setting
- **WhatsApp notification toggle:** Per-subscription setting

## Sentiment Analysis (Local, No AI)

Positive words (+1 each): wonderful, professional, best, great, recommend, excellent, love, good, quality, friendly, helpful. Negative words (-1.2 each): terrible, waited, frustrating, disappointed, expensive, rude, dirty, late, worst, bad, awful, slow.

```
Score > 0.5 = POSITIVE | Score < -0.5 = NEGATIVE | Otherwise = NEUTRAL
```

# 9. AI Service (Gemini)

## Configuration

```
GEMINI_API_KEY, GEMINI_MODEL=gemini-3.1-flash-lite, GEMINI_IMAGE_MODEL=gemini-2.5-flash-image
```

## Cost Optimizations Applied

1. **Cheapest model:** gemini-3.1-flash-lite ($0.25/$1.50 per million tokens)
2. **Thinking disabled:** thinkingConfig => ['thinkingBudget' => 0] in all calls
3. **Per-call token limits:** Review reply=150, Content=200, Keywords=300, Chat=400
4. **Local sentiment:** No AI call for sentiment analysis at all
5. **Short prompts:** All prompts condensed, no persona wrapper for JSON responses

## Public Methods

| Method | Max Tokens | Use Case |
|---|---|---|
| analyzeSentiment() | 0 (no AI) | Local keyword matching |
| generateReviewReply() | 150 | AI review reply with fallback |
| generateContent() | 200 | Social media post copy |

| Method | Max Tokens | Use Case |
|---|---|---|
| rawPrompt() | varies | Direct prompt, JSON responses |
| chat() | 400 | AI marketing assistant |
| generateImage() | N/A | Image generation (separate model) |
| generateKeywords() | 300 | SEO keyword suggestions |

### Fallback Pattern

Every AI method has a local/hardcoded fallback. The app works fully without any API key - all AI responses gracefully degrade to static content.

# 10. GBP Content (Posts & Photos)

### Posts

CRUD + Publish to Google via GBP API. AI-generated post body (credit-gated). Statuses: DRAFT, SCHEDULED, PUBLISHED. Fields: gbp_location_id, type, body, image, cta_url, status, scheduled_at, published_at.

### Photos

Upload + Publish to Google. AI caption generation (credit-gated). Fields: gbp_location_id, image, caption, status, scheduled_at.

Public image endpoints (outside auth) for Google servers to fetch sourceUrl.

# 11. Social Media Publishing

### Platforms

| Platform | Status | Details |
|---|---|---|
| Facebook | Full API | Text + image via Meta Graph API |
| Instagram | Full API | Image required, two-step container flow |
| LinkedIn | Draft only | No API integration |
| X (Twitter) | Draft only | No API integration |

Meta OAuth connects via /clients/{client}/meta/connect. Scopes: pages_show_list, pages_manage_posts, pages_read_engagement, instagram_basic, instagram_content_publish, business_management.

Features: Post to multiple platforms, AI caption generation, retry failed posts, draft-to-publish workflow.

# 12. Lead Management (CRM)

### Kanban Stages

```
NEW -> CONTACTED -> QUALIFIED -> CONVERTED -> LOST

Fields: agency_id, client_id, name, phone, email, source, stage, score, notes
```

Features: Create leads manually, move between stages (drag/drop kanban), delete leads, filter by stage/source.

# 13. WhatsApp Integration

Uses Meta's WhatsApp Cloud API. Send templated or free-form messages. Track message status.

```
Fields: agency_id, to_phone, from_phone, direction (OUTBOUND/INBOUND), template, body, status
```

# 14. Ad Reports

Manual report entry (no API integration currently). View campaign performance metrics. Filter by network/client/period.

```
Fields: agency_id, client_id, network (GOOGLE_ADS/META_ADS), campaign, spend, clicks, impressions,
conversions, leads, period_start, period_end
```

# 15. Business Insights

Live Google performance metrics via GBP API. Download as CSV. Metrics: views, searches, calls, direction requests, website clicks.

# 16. Google Audit

### Scoring (0-100)

| Check | Weight | Pass Condition |
|---|---|---|
| Google location connected | 15 | Has at least 1 location |
| Reviews collected | 15 | Total reviews > 0 |
| Healthy average rating | 20 | Average >= 4.0 (partial: >= 3.0) |
| Replying to reviews | 25 | Reply rate >= 80% (partial: >= 50%) |
| Negative reviews addressed | 25 | All negative reviews replied |

Grades: >= 85 Excellent | >= 65 Good | >= 40 Needs work | < 40 Poor

AI generates 3 bullet-point tips (150 tokens). Requires active plan + 2 credits. Hardcoded fallback if AI unavailable.

# 17. Competitor Analysis

User enters up to 5 competitor names + city. System gathers review stats. AI compares and returns JSON: summary, strengths, gaps, actions. Requires active plan + 2 credits.

# 18. Local Rank Checker

**Real mode (Google Places API):** Searches top 20 results for business name match. Returns rank, checked count, result list.

**Fallback mode (AI Estimate):** Returns estimated_rank, summary, tips. Requires active plan + 1 credit. Places API key is separate from OAuth.

# 19. Keyword Generator

User enters business name, city, optional industry. AI returns 4 keyword groups: High intent, Local, Long-tail, Services (5-7 keywords each). Requires active plan + 2 credits. Fallback generates generic keywords.

# 20. One-Click Optimization

Analyzes current business profile state. Runs AI-powered optimization suggestions. Credit-gated: 3 credits per run.

# 21. AI Chat Mode

Full-page marketing assistant chat. Context: Google reviews, social media, local SEO, marketing advice. 1 credit per message, max 400 output tokens. Requires active plan.

# 22. AI Media Generation

Text-to-image generation via Gemini (gemini-2.5-flash-image model). Images stored as base64 data URIs. 'Use as Photo' converts AI media to GBP photo. 5 credits per generation (most expensive). Fallback: SVG placeholder.

# 23. Invoicing & Accounting

| Feature | Description |
|---|---|
| Invoices | Create, view, mark sent, mark paid, delete |
| Customers | CRUD (uses Client model) |
| Services | CRUD with categories and GST per service |
| Service Categories | Group services for organization |
| Expenses | Track business expenses |
| Billing Settings | Company info, GSTIN, bank details, invoice prefix |
| Tally Export | Export data in Tally-compatible format |

GST Features (India-specific): Per-service GST percentage, per-plan GST rate, GST-inclusive pricing with base/GST split, GSTIN stored on Client and BillingSetting, Tally export for accounting.

# 24. Team Management

List team members (agency-scoped), invite new members, update role/client assignment, remove members. Only CLIENT_OWNER can manage team. Members can be assigned to specific clients.

# 25. Admin Panel (Super Admin)

| Feature | Description |
|---|---|
| Overview Dashboard | System-wide stats |
| User Management | CRUD users across all agencies |
| Plan Management | CRUD plans (name, price, GST, credits, features, permissions) |
| Credit Packages | CRUD packages, toggle active/inactive |
| Billing/Payments | View all payments, checkout, verify |
| Manual Top-up | Add credits to any agency |

Protected by middleware: auth + role:SUPER_ADMIN. Admin bypasses EnsureActivePlan middleware.

# 26. Health Score Dashboard

| Category | What's Measured | Target |
|---|---|---|
| Reviews | Rating + reply rate | 4.0+ avg, 80%+ replies |
| Google Posts | Published vs total | Regular publishing |
| Photos | Photo count | 10+ photos |
| Social | Published vs draft | Publish drafts |
| Leads | Conversion rate | Follow up promptly |
| Connections | Google + Meta connected | Both connected |

Overall score: average of all 6 sub-scores (0-100). Includes prioritized 'Do Next' actions list (up to 6 items) with impact levels: critical, high, medium, low.

# 27. Email Verification

Send verification email (signed URL), verify via signed link (no auth required), change email address. Email verification required before connecting Google OAuth. Uses Laravel's signed URL mechanism.

# 28. Mobile API (Flutter)

Auth: Laravel Sanctum Personal Access Tokens (Authorization: Bearer {token}).

**106+ routes** across **18 API controllers** in app/Http/Controllers/Api/. All CRUD endpoints mirror web controllers with same agency_id scoping and credit gating, returning JSON responses only.

## API Controllers

```
AuthController, DashboardController, ClientController, ReviewController, GbpContentController,
LeadController, SocialController, BillingController, WhatsappController, AdController,
InsightsController, TeamController, AuditController, CompetitorController, KeywordController,
RankCheckerController, OptimizationController, AiModeController, AiMediaController, AdminController,
InvoicingController
```

# 29. Environment Variables

| Variable | Purpose | Required |
|---|---|---|
| GEMINI_API_KEY | Google Gemini AI text/image generation | For AI features |
| GEMINI_MODEL | AI model (default: gemini-3.1-flash-lite) | No |
| RAZORPAY_KEY_ID | Razorpay payment gateway key | For payments |
| RAZORPAY_KEY_SECRET | Razorpay secret (HMAC verification) | For payments |
| GOOGLE_CLIENT_ID | Google OAuth (Business Profile) | For GBP |
| GOOGLE_CLIENT_SECRET | Google OAuth secret | For GBP |
| GOOGLE_PLACES_API_KEY | Google Places (Rank Checker) | For rank check |
| GOOGLE_LOGIN_CLIENT_ID | Google login (separate from GBP) | For Google login |
| GOOGLE_MOBILE_CLIENT_ID | Google login for Flutter app | For mobile |
| META_APP_ID | Meta/Facebook/Instagram app | For social |
| META_APP_SECRET | Meta app secret | For social |
| WHATSAPP_PHONE_NUMBER_ID | WhatsApp Cloud API phone | For WhatsApp |
| WHATSAPP_ACCESS_TOKEN | WhatsApp Cloud API token | For WhatsApp |

# 30. Database Schema (All 24 Models)

| Model | Table | Key Relationships |
|---|---|---|
| Agency | agencies | hasMany: Users, Clients. hasOne: Subscription |
| User | users | belongsTo: Agency, Client. HasApiTokens, Notifiable |
| Client | clients | belongsTo: Agency. hasMany: Locations, Leads, Invoices |
| Subscription | subscriptions | belongsTo: Agency. Tracks plan, credits, status |
| Plan | plans | Plan definitions (price, GST, features, permissions) |
| GbpLocation | gbp_locations | belongsTo: Client. hasMany: Reviews, Posts, Photos |
| Review | reviews | belongsTo: GbpLocation. sentiment, reply, draft_reply |
| GbpPost | gbp_posts | belongsTo: GbpLocation. DRAFT/SCHEDULED/PUBLISHED |
| GbpPhoto | gbp_photos | belongsTo: GbpLocation |
| Integration | integrations | OAuth tokens (GOOGLE_GBP, META_GRAPH) |
| SocialPost | social_posts | Multi-platform. media_urls (JSON) |
| Lead | leads | Kanban: NEW/CONTACTED/QUALIFIED/CONVERTED/LOST |
| WhatsappMessage | whatsapp_messages | Direction: OUTBOUND/INBOUND |
| AdReport | ad_reports | Campaign metrics with date range |
| Payment | payments | Razorpay order + payment tracking |
| CreditLedger | credit_ledger | Audit trail of credit changes |
| CreditPackage | credit_packages | Buyable credit packs with GST |
| AiMedia | ai_media | AI-generated images (base64) |
| Invoice | invoices | DRAFT/SENT/PAID. GST split |
| InvoiceItem | invoice_items | Line items with per-item GST |
| Service | services | Billable services with categories |
| ServiceCategory | service_categories | Group services |
| Expense | expenses | Business expense tracking |
| BillingSetting | billing_settings | Company info, bank, GSTIN, Tally |

# Appendix: Route Groups & Key Patterns

## Route Access Levels

| Access Level | Features |
|---|---|
| Public (no auth) | Landing page, policy pages, OAuth callbacks, email verification link, media endpoints |
| Auth (free) | Dashboard, Clients CRUD, Reviews (view/sync/reply), GBP Content, Leads, Social, WhatsApp, Ads, Insights, Invoicing, Team, Profile, Billing/Plans |
| Auth + Plan (AI) | AI review replies, AI captions, One-click optimize, Keyword generation, Rank checker, AI media, Google audit, Competitor analysis, AI chat, Buy credits |
| Auth + SUPER_ADMIN | Admin overview, User management, Plan management, Credit packages, Billing management, Manual top-up |

## Key Patterns to Replicate

- **1. Multi-Tenancy via agency_id:** Every model scoped to an agency. Every controller reads $request->user()->agency_id.
- **2. Credit Gating:** canAfford() - do work - deduct(). Transactional with row locking.
- **3. AI with Graceful Fallback:** Every AI call has a static/hardcoded fallback. App works without API keys.
- **4. Dual Interface (Web + API):** Web controllers return Blade views. API controllers return JSON. Same business logic.
- **5. OAuth State in Cache:** Google/Meta OAuth state stored in Cache (not session) to survive cross-domain redirects.
- **6. Service Provider Pattern:** GbpService - GoogleGbpProvider (real) or MockGbpProvider (fake). Controllers never know which.
- **7. GST-Inclusive Pricing:** All prices stored as GST-inclusive. Base and GST amounts computed on the fly.
- **8. Middleware Layering:** auth = base features (free) | auth + plan = AI features (paid) | auth + role:SUPER_ADMIN = admin.